

The vertex / MPM interface:

a surface and a continuum exchanging force, and the spheroid it was built for

prototype/ecm/mesh_contact_ops.py · test_03_mesh_contact.py · test_04_spheroid_ecm.py · gates_03.py ·
log/okuda_ECM/03*, 03d_gates, 04* · 10 August 2026

1. Mechanism

An epithelium pressing into a matrix is two materials touching: a sheet of cells that is a *surface* with no thickness worth resolving, and a stroma that is a *continuum*. Everything biological that happens between them — adhesion, contact, load transmission, the slip that lets a tissue slide over a gel instead of welding to it — crosses that boundary, and it is a boundary between two different kinds of object rather than between two materials.

The obvious numerical scheme does not fit, and the reason is a ratio. Outside biology this is FEM-MPM coupling, and its natural form is *grid-node* coupling (CFEMP [1]): resolve contact by comparing the two bodies' velocities at grid nodes they share. It requires the mesh and the grid to be comparable in size. Ours are not. A cell of this epithelium is 0.27 to 0.42 grid cells across and a basement membrane is 0.1, so both bodies fall inside one cell, the grid hands them a single mass-weighted velocity, and they are *welded* — which is what runs 142–151 measured ten times over. *Particle-to-surface* coupling (ICFEMP [2]) removes the restriction by testing a particle against a mesh *face* rather than against a node, and that is the mechanism here.

What the mechanism has to carry, biologically. Three things, and each is a term in §2. A *normal* force, so the matrix is displaced rather than passed through. Its *reaction*, so the tissue feels what it pushes — a one-way push is a decorative deformation, and the whole point of a matrix that pushes back is that confinement becomes an outcome instead of a boundary condition. And *friction*, because a real epithelium slides against a gel: with no tangential law the shared grid gives no-slip for free, and a "contact" that is MPM's automatic weld wearing a coefficient is not a contact.

2. Equations and symbols

For particle p under face f , let $\hat{\mathbf{n}}$ be the face's outward normal and $\mathbf{w}$ the barycentric weights of p 's contact point on it. The penetration, the normal penalty and the regularised Coulomb friction are

$$d_p = [(\mathbf{x}_f - \mathbf{x}_p) \cdot \hat{\mathbf{n}}]_+, \quad \mathbf{f}_p^n = k d_p \hat{\mathbf{n}}, \quad (1)$$

$$\Delta \mathbf{v}_t = (\mathbf{v}_p - \sum_i w_i \mathbf{v}_i)_{\perp \hat{\mathbf{n}}}, \quad \mathbf{f}_p^t = -\min\left(\mu k d_p, \mu k d_p \frac{|\Delta \mathbf{v}_t|}{\varepsilon}\right) \frac{\Delta \mathbf{v}_t}{|\Delta \mathbf{v}_t|}, \quad (2)$$

and the reaction is distributed to the face's vertices by the same weights, which is what makes the exchange conservative by construction:

$$\mathbf{f}_i = -w_i (\mathbf{f}_p^n + \mathbf{f}_p^t), \quad \text{so} \quad \sum_p \mathbf{f}_p + \sum_i \mathbf{f}_i = \mathbf{0}. \quad (3)$$

ε is a slip velocity and not a fudge: it is the speed below which a contact is treated as stuck, so a resting contact does not chatter.

Finding the face is a ray, and that is what makes it exact. The tissue is star-shaped about its own centroid, so the contact point is the intersection of the ray from the centroid through the particle with the surface. Möller–Trumbore on triangle $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ gives the weights and the surface radius t in one solve:

$$t = \frac{\mathbf{e}_2 \cdot \mathbf{q}}{\mathbf{e}_1 \cdot \mathbf{p}}, \quad w_1 = \frac{\mathbf{s} \cdot \mathbf{p}}{\mathbf{e}_1 \cdot \mathbf{p}}, \quad w_2 = \frac{\hat{\mathbf{u}} \cdot \mathbf{q}}{\mathbf{e}_1 \cdot \mathbf{p}}, \quad \mathbf{x}_f = \mathbf{c} + t \hat{\mathbf{u}}_p, \quad (4)$$

with $\mathbf{e}_{1,2}$ the triangle's edges from $\mathbf{A}$, $\mathbf{s} = \mathbf{c} - \mathbf{A}$, $\mathbf{p} = \hat{\mathbf{u}} \times \mathbf{e}_2$, $\mathbf{q} = \mathbf{s} \times \mathbf{e}_1$. t is the curved-surface form of a flat patch's interpolated height, and the weights fall out of the same solve rather than being a second, separate projection.

The one constraint that is not optional. The contact reaches the particle as a body force integrated at Δt , so it is stable only while $\Delta t \sqrt{k/m} < 1$:

$$k < k_{\max} = \frac{m}{\Delta t^2}, \quad \text{written as} \quad k = (\mathbf{k_frac}/\Delta t)^2 \text{ in acceleration units.} \quad (5)$$

The first version of the flat rig used $k = 3 \times 10^4$ against a ceiling of 95, three hundred times past it, and the same mistake NaN'd the flat sheet at $k = 6 \times 10^3$. Writing the stiffness as a *fraction* of its own ceiling is what stops it being made a third time.

symbol	is	of what
d_p	penetration	of particle p behind the face, along the face NORMAL, not along the ray
k	penalty stiffness	in acceleration units; $k_{\text{frac}}^2/\Delta t_{\text{sub}}^2$, and $k_{\text{frac}} = 0.15$ throughout
μ	friction coefficient	Coulomb, regularised below ε
ε	slip velocity	the speed below which a contact is stuck; 10^{-3}
$\mathbf{w}$	barycentric weights	of the contact point; they build the force AND distribute its reaction — one array, so the two cannot disagree
t	surface radius along the ray	Eq. (4); the star-shape’s stand-in for a height
$\hat{\mathbf{u}}_p$	particle direction	from the tissue centroid, which is also the bin key
$\mathbf{a}_{\text{max}}$	the operator’s own clamp	applied BEFORE the reaction is recorded, so the momentum gate cannot pass on a force the scatter later clipped

3. The Plexus2 container

Contact is a relation, so it is an edge set and not a force field on a body.

set	kind	state it carries
contact	edge, pre: mpm_particle, post: vertex	d_p , $\mathbf{w}$, the face, bound flag
surface	node set, one per membrane node	direction and radius — <i>deleted by this operator</i> , see below

operator	kind	acts on	mechanism	its gate
contact_rebuild	Rewire	contact	which particle is under which face; the (θ, ϕ) bin lookup	S1–S3
mesh_contact	Lateral	contact	Eqs. (1)–(3); returns a delta to <i>both</i> endpoints in one call	S4–S8
mesh_inside_count	Lateral (diagnostic)	mpm_particle	counts what is behind the surface; does not move it	S7

The lookup, which is the only thing a curved surface needs that a flat one does not. A direction names the face. The bin structure is the one `apical_map` already uses for $R(\theta, \phi)$, holding the *triangles* instead of a radius — and that difference is the operator: a bin of the map is a NUMBER, so it can only push radially and can return its reaction to nothing, while a bin here holds three real vertices. Two details are not free. **The grid is re-sized every frame**, because the nine-bin query is exact only while a triangle fits inside a bin and the faces shrink from 0.17 to 0.028 rad as the tissue divides: 82 bins in 8 rows at frame 0, 3576 in 53 rows at the last, at most 64 triangles in a bucket. And **the rows are reduced** — each given as many ϕ bins as it can hold *square* ones, $n_\phi = 2\pi \sin \theta / d\theta$ — because on a plain (θ, ϕ) lattice a bin near the pole is a sliver, a triangle spans ten of them, and the query misses the face the particle is standing on: silently, only near the poles, with matrix inside the tissue as the symptom.

Placement, and it is worth $64\times$. `mesh_contact` goes *inside* the `substep_dt` block, recomputed at the substep’s own positions; at frame level the ceiling in Eq. (5) falls by $(\Delta t_{\text{frame}}/\Delta t_{\text{sub}})^2$. The diagnostics go after it, at frame level, because a non-penetration count must test this frame’s positions against *this* frame’s mesh.

```

schedule:
- aggregate -> seed_ecm
- {substep_dt: 4.0e-4, steps: [mesh_contact, mpm_strain, mpm_scatter,
                               mpm_grid_update, mpm_gather]}
- ecm_stress -> mesh_inside_count

```

What it deletes. `cell_to_ecm[replay]`, a radial force read off a smoothed 32×64 radius map — it cannot return a reaction (a bin has no vertices) and cannot generate a shear (a radial force has no tangential part). And `cell_exclude_3d`, the projection backstop, because it makes non-penetration *unmeasurable*: with it on, the answer is zero by construction.

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. S1–S3 run before any physics, as `mesh_contact_ops.selftest()`; a run refuses to start if they fail. S12–S16 are the sweep no penalty method may skip: the stiffness, the time step and the grid spacing are not physics, and a result that moves when they move is a result about the discretisation.

gate		threshold
<i>the lookup, before any physics</i> (meshes 0, 100, 199)		
S1	a ray from the centroid hits a closed star-shaped surface	coverage 1.00000 on all three > 0.9999
S2	the binned face is the one a test against EVERY triangle finds	0 0 , max relative 0.0 of 400 rays
S3	bucket occupancy stays bounded as the mesh refines	< 64 at 38,268 triangles 128 per bucket
<i>the interface</i> (03, 03b, 03c; 04, 04c/d)		
S4	momentum: $ \sum \mathbf{f}_{\text{part}} + \sum \mathbf{f}_{\text{vert}} / \sum \mathbf{f} $, flat patch	< 1.5e-7 (float32 precision) 10^{-6}
S5	the same on a curved, dividing, re-meshed surface	< 1.2e-7 max, 1.5e-8 median 10^{-6}
S6	penetration stays bounded	< 1.47 (03b), 0.82 (04), 0.73 (04d) 2 cells
S7	particles behind the surface are the CONTACT, not a leak	ratio 0.90–0.96 ; deepest 0.90R , 0 below $0.7R$ to par- ti- cles in con- tact < 1.5; none be- low $0.7R$
S8	friction changes the slip: $\mu = 0.4$ against $\mu = 0$	> 5.3 × (03b), 3.9 × (03c) 2×
S9	the operator's own clamp never binds	max 1.8e3 against 3×10^4 a_max
<i>what the spheroid does to the matrix</i> (04, 04c, 04d)		
S10	the far field is the bounded shell's own $C_1 r + C_2 / r^2$	R^2 > 0.94–0.999 (04, S/N 3.9–5.5); 0.88–0.998 (04c/d, S/N 1.9–2.9) 0.90
S11	strand reorientation is its own affine control	with 10.2725 vs 0.2714 5%
<i>the discretisation</i> (03d_gates, 1600 frames, k over $16\times$)		
S12	momentum residual in EVERY variant	< 1.47–1.57e-7 10^{-6}
S13	penetration scales as $1/k$	log- -0.15 ($1.92 \rightarrow 1.27$ cells) log slope in [−1.4, −0.6] spread 1.119 ($13.88 \rightarrow 14.95 \rightarrow 15.66\%$) < 0.10 of nom- i- nal
S14b	the transmitted impulse does not move with k	spread 1.181 < 0.15 of nom- i- nal
S15	the answer is converged in the time step	compression 0.935 ($14.95 \rightarrow 18.50\%$ at half the step) within 0.05
S16	the interface does not consult the grid: $n_{\text{grid}} 64 \rightarrow 96$	compression 0.938 , depth 0.067 within 0.15, depth (box) within 0.5
<i>is the failure rate-limited?</i> (03e_press_rate, press over $8\times$ at matched)		
S17	the answer converges as the press slows	travel) change 0.011 ($19.49 \rightarrow 14.95 \rightarrow 15.44 \rightarrow 15.61\%$) be- tween the two slow- est < 0.05 spread 1.136 ($14.19/15.61/16.31\%$) — it does NOT < 0.10 (it is 0.119 at nom- i- nal)
S18	the k -dependence weakens when the rate falls	3
S19	friction changes the slip on a CURVED, moving surface	> 1.73 × (5.22 vs 3.02×10^{-2} , 04e) 2×,

The run declares its scale, and these numbers are what it makes of it. Three base scales under `general.units`: and everything else derived (`plexus/units.py`): 1 box unit = $1172\ \mu\text{m}$, because a cell is $8.5\ \mu\text{m}$ and the tissue is solved at $10\ \mu\text{m}$ per tissue unit; 1 frame = 600 s, because $200 \rightarrow 6,076$ cells is 4.93 doublings and a 12–24 h cycle puts the run at 2.5–4.9 days; 1 force unit = 9162 nN, set so the stroma’s `youngs`: 15 is 100 Pa — a collagen I gel at 1–3 mg/ml. Only the force scale is *chosen*, and it is chosen on the softest and best-measured body rather than on the sheet under dispute. Without this block the loader says the run is dimensionless and no result from it may carry a unit, which was the state of every run in this folder until now.

Two conventions make these mean what they say. The momentum residual is taken on the contact’s own force pair, inside the substep and before anything else is added — with a drive in the sum it stops being a statement about the reaction. And **penetration is reported in box units as well as in cells** across the grid sweep, because dx is exactly what changes there and cells alone would report a grid dependence that is the unit.

S13–S15 are the finding of this note, and S17–S18 are what became of the explanation. A penalty permits a depth $d = f/k$, so a sixteenfold stiffening should divide the penetration by sixteen and leave the material’s compression alone. Measured, the penetration falls by $1.5\times$ and the compression *rises* by 13%; halving the time step moves it again, $14.95 \rightarrow 18.50\%$.

The obvious explanation was that the contact is rate-limited, and 03e rejected it. The surface is prescribed, so it advances $v\Delta t$ per step whatever the matrix does; if the penalty cannot accelerate a particle out of the way within a step, the depth is set by the advance rather than by the force, and all three sweeps become sweeps of how much advance happens per step. That is testable, because it predicts two things. The first holds: pressing at 2, 1, $\frac{1}{2}$ and $\frac{1}{4}$ of the nominal speed at matched total travel, the compression converges — $19.49 \rightarrow 14.95 \rightarrow 15.44 \rightarrow 15.61\%$, the last two within 1.1% of each other (S17), so the nominal press is already converged to 4% and the time-step failure S15 is a resolution question rather than a mechanism. **The second fails.** At the slowest press the k -dependence does not weaken; it is 13.6% over a $16\times$ change in k against 11.9% at the nominal (S18). A rate-limited contact must lose its k -dependence when the rate falls, and this one does not — so the compliance is intrinsic to the penalty and not an artefact of pressing too fast, and the answer is the barrier formulation (BFEMP [3]) rather than a slower press. That is a prediction made before the run and refuted by it, which is the only reason to have made it.

S19 is the same lesson in the other direction. Friction changes the slip four- to fivefold on the flat patch, and only $1.73\times$ on the spheroid — because the spheroid’s loading is very nearly pure normal expansion, so there is little tangential relative motion for a friction law to act on. The threshold was set on the flat rig and does not transfer: what 04e shows is not that the friction law is broken but that the curved run does not exercise it, and a tangential gate for the spheroid needs a loading with shear in it.

What survives all of this is S12, and S16 with it: the reaction is exact at $1.5\text{--}1.8 \times 10^{-7}$ in every variant of every sweep — four press speeds, three stiffnesses, two time steps, two grids, two friction coefficients — because Eq. (3) is an identity on $\mathbf{w}$ and not a balance that has to converge. And refining the grid at fixed particle count moves the compression by 12% and the penetration in box units by 7%, which is the material’s response and not the interface’s: the one place the scheme’s own claim, that particle-to-surface contact never consults the grid, is tested and holds.

What the gates license. A surface and a continuum exchange force conservatively at a resolution where grid-node coupling welds them (S4, S5); the slip is a material property rather than a numerical accident (S8); the non-penetration the method fails to achieve is *measured* rather than projected away (S6, S7); and the matrix responds as the isotropic elastic shell it was given (S10, S11). **What they do not license:** the quantitative contact — S13 and S14 fail, so any number that depends on how hard the surface pressed is provisional until the penalty is replaced by a barrier formulation (BFEMP [3], which forbids penetration outright at the cost of an implicit solve) or the press is made force-driven rather than displacement-driven. Nor two-way mechanics: the tissue is still a replay, so the reaction it computes is recorded and dropped. Handing `vertex_force` to a live epithelium is a schedule change rather than an architecture change, and it is the next run.

5. References

References

- [1] Lian, Y.P., Zhang, X., Liu, Y. (2011) Coupling of finite element method with material point method by local multi-mesh contact method. *Comput. Methods Appl. Mech. Eng.* **200**:3482–3494.
- [2] Chen, Z.-P., Qiu, X.-M., Zhang, X., Lian, Y.-P. (2015) Improved coupling of finite element method with material point method based on a particle-to-surface contact algorithm. *Comput. Methods Appl. Mech. Eng.* **293**:1–19.
- [3] Li, X., Fang, Y., Li, M., Jiang, C. (2022) BFEMP: Interpenetration-free MPM–FEM coupling with barrier contact. *Comput. Methods Appl. Mech. Eng.* **390**:114350.
- [4] Möller, T., Trumbore, B. (1997) Fast, minimum storage ray/triangle intersection. *J. Graphics Tools* **2**(1):21–28.

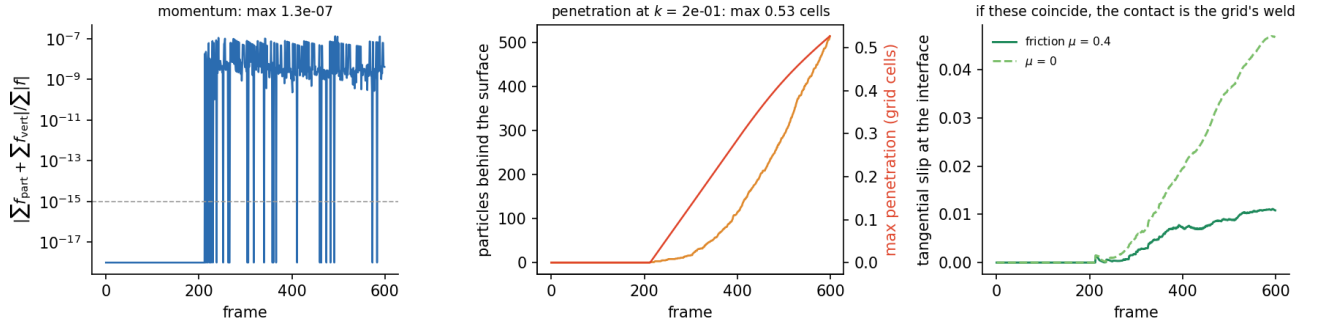


Figure 1: 03, the flat rig. **a** the momentum residual (S4), which a one-way coupling fails by construction. **b** how many particles are behind the surface and how deep, in grid cells (S6). **c** tangential slip with friction and without (S8) — if these coincided, the contact would be MPM’s automatic weld wearing a friction coefficient.

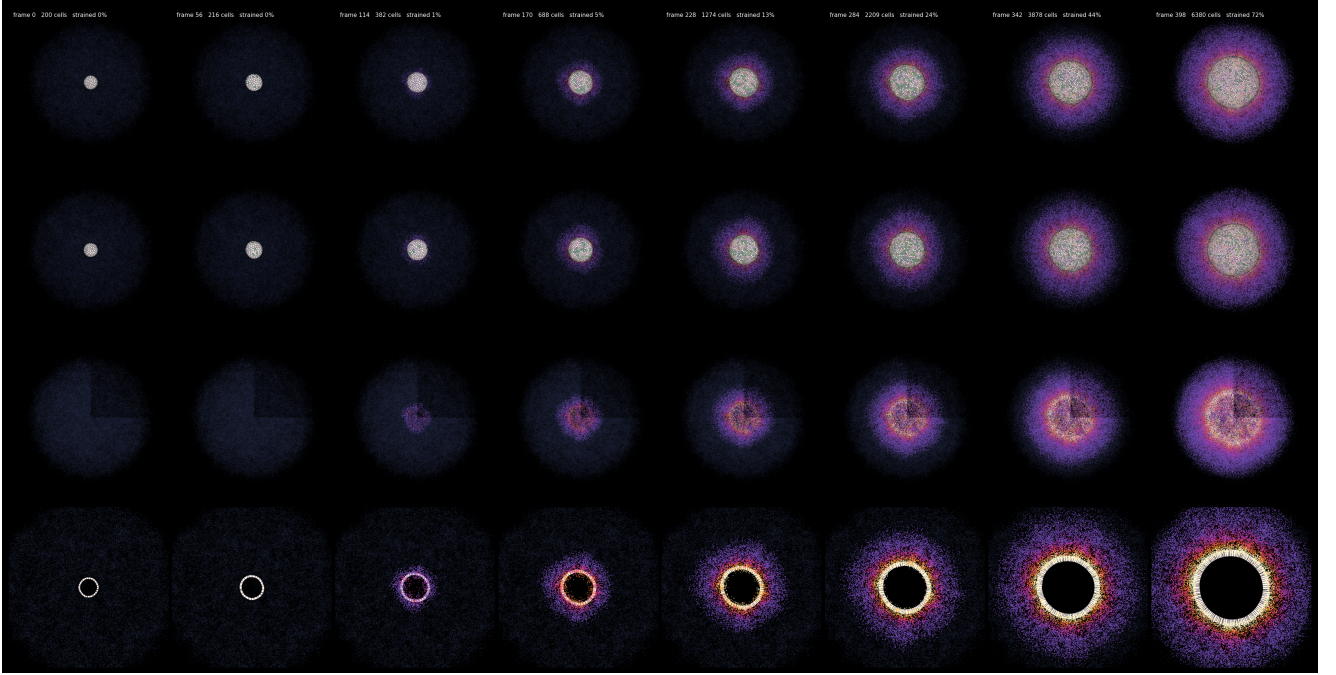


Figure 2: 04d, the section, coloured by von Mises stress on one scale over the whole run. The spheroid clears a hole in the matrix and a shell of compacted, stressed strands travels outward ahead of it while the far field stays unloaded; the matrix nearest the tissue reaches 1.25 times its own seeded density at $r = 0.158$, just outside the surface. First contact is frame 32 of 400, where the cavity radius and the tissue radius cross.

- [5] Okuda, S., Miura, T., Inoue, Y., Adachi, T., Eiraku, M. (2018) Combining Turing and 3D vertex models reproduces autonomous multicellular morphogenesis. *Sci. Rep.* **8**:2386.
- [6] Hu, Y. *et al.* (2018) A moving least squares material point method with displacement discontinuity and two-way rigid body coupling. *ACM Trans. Graph.* **37**(4):150.
- [7] Wang, H., Abhilash, A.S., Chen, C.S., Wells, R.G., Shenoy, V.B. (2014) Long-range force transmission in fibrous matrices enabled by tension-driven alignment of fibers. *Biophys. J.* **107**(11):2592–2603.